Orion S. Ning

orion.ning@berkeley.edu | Personal Website | **Github**: orionning676 | Updated October, 2026

Academic Positions

PhD Student, Theoretical Physics 2022 –
University of California, Berkeley, and Lawrence Berkeley National Laboratory
NSF Graduate Research Fellow

Education

University of California, Berkeley Berkeley, CA
Ph.D. Physics, Advisor: Benjamin R. Safdi *2022 –*

The University of California, Berkeley Berkeley, CA
B.A. Physics (Highest Honors), Mathematics (High Honors) *2022*
M.A. Physics *2024*

Current Research Interests

Dark Matter Phenomenology and Indirect Detection, Axions/ALPs, Gravitational Waves, Astroparticle Physics, Gamma-Ray Astronomy, Computational Methods

Publications

INSPIRE |

[1] Robin Anthony-Petersen, Orion Ning, Yu Watanabe, Benjamin R. Safdi, Maíra Dutra, et al. "MeV dark matter decay forecasts for the COSI mission". Submitted to *Phys. Rev. D*. 2026. arXiv: 2610.17217 [astro-ph.CO].

[2] Orion Ning, Benjamin R. Safdi, and Catherine Welch. "No Evidence for Superradiant Axions in LIGO-Virgo-KAGRA GWTC-5 Binary Black Hole Spins". Submitted to *Phys. Rev. Lett.* 2026. arXiv: 2607.01317 [hep-ph].

[3] Sebastian A. R. Ellis, Orion Ning, Nicholas L. Rodd, and Jan Schütte-Engel. "Crowdsourcing Gravitational Waves from Superradiant Axions". Accepted to *Phys. Rev. D*. 2026. arXiv: 2603.15734 [hep-ph].

[4] Andrea Giovanni De Marchi, Orion Ning, and Tianzhuo Xiao. "Blazar Constraints on Axions through New Spectral Modulation Searches in 1ES 1959+650 & B2 1811+31". Submitted to *Phys. Rev. D*. 2026. arXiv: 2603.13480 [hep-ph].

[5] Orion Ning, Kailash Raman, and Benjamin R. Safdi. "Cosmological Neutron Stars Produce Diffuse Axion X-Ray Signatures". Submitted to *Phys. Rev. Lett.* 2025. arXiv: 2512.15849 [hep-ph].

[6] Orion Ning, Anupam Ray, and Benjamin R. Safdi. "Axion lines from nuclear de-excitations in galactic stellar populations". In: *Phys. Rev. D* 113.3 (2026), p. 035010. DOI: 10.1103/xyjw-9bw7. arXiv: 2509.03569 [hep-ph].

[7] Orion Ning and Benjamin R. Safdi. "Probing the axion-electron coupling with NuSTAR observations of galaxies". In: *Phys. Rev. D* 113.1 (2026), p. 015021. DOI: 10.1103/28vj-3rws. arXiv: 2503.09682 [hep-ph].

[8] Orion Ning, Christopher Dessert, Vi Hong, and Benjamin R. Safdi. "Search for axions from magnetic white dwarfs with Chandra x-ray observations". In: *Phys. Rev. D* 111.10 (2025), p. 103002. DOI: 10.1103/PhysRevD.111.103002. arXiv: 2411.05041 [astro-ph.HE].

[9] Orion Ning and Benjamin R. Safdi. "Leading Axion-Photon Sensitivity with NuSTAR Observations of M82 and M87". In: *Phys. Rev. Lett.* 134.17 (2025), p. 171003. DOI: 10.1103/PhysRevLett.134.171003. arXiv: 2404.14476 [hep-ph]. **Editors' Suggestion**.

[10] Christopher Dessert, Orion Ning, Nicholas L. Rodd, and Benjamin R. Safdi. "Resurrecting Hitomi for Decaying Dark Matter and Forecasting Leading Sensitivity for XRISM". In: *Phys. Rev. Lett.* 132.21 (2024), p. 211002. DOI: 10.1103/PhysRevLett.132.211002. arXiv: 2305.17160 [astro-ph.CO].

[11] K. Byrum et al. (Mu2e-II Collaboration, incl. O. Ning). "Mu2e-II: Muon to electron conversion with PIP-II". In: *Snowmass 2021*. 2022. arXiv: 2203.07569 [hep-ex].

[12] Cristhian Garcia-Quintero, Mustapha Ishak, and Orion Ning. "Current constraints on deviations from General Relativity using binning in redshift and scale". In: *JCAP* 12 (2020), p. 018. DOI: 10.1088/1475-7516/2020/12/018. arXiv: 2010.12519 [astro-ph.CO].

Selected Talks and Presentations

- "Crowdsourcing Gravitational Waves from Superradiance: Black Holes as Extreme Axion Laboratories," N3AS Seminar, *virtual, September 2026*
- "Crowdsourcing Gravitational Waves from Superradiance: Black Holes as Extreme Axion Laboratories," PPC 2026, *Sydney, Australia, September 2026*
- "Galactic Axion Factories: Astrophysical Sources and Signals for Axion Searches," King's College London TPPC Seminar, *London, UK, July 2026*
- "Dark Particles and Bright Bursts: Future Stories from the Gamma-Ray Sky with GALAXIS," Closing In on Thermal Relics with Indirect, Direct, and Collider Searches for Dark Matter Conference (CERN), *Geneva, Switzerland, July 2026*
- "Axion Searches from Black Hole Superradiance," CERN Theory Cosmo/BSM Coffee, *Geneva, Switzerland, July 2026*
- "Crowdsourcing Gravitational Waves from Superradiance: Black Holes as Extreme Axion Laboratories," 4th General Meeting of COST Action COSMIC WISPers CA21106, *Corfu, Greece, June 2026*
- "Dark Particles and Bright Bursts: Future Stories from the Gamma-Ray Sky with GALAXIS," COST WG3 (WISPs in Astrophysics) Monthly Meeting, *virtual, April 2026*
- "Dark Particles and Bright Bursts: Future Stories from the Gamma-Ray Sky with GALAXIS," LBL Dark Matter Lunch Talk, *Berkeley, CA, April 2026*
- "Galactic Axion Factories: Astrophysical Populations and Signals for Axion Searches," Stanford KIPAC Seminar, *Stanford, CA, March 2026*
- "Galactic Axion Factories: Stellar Populations and X-Ray Signals for Axion Searches," SLAC Theory Seminar, *Stanford, CA, October 2025*
- "Dark Particles and Bright Bursts: Future Stories from the Gamma-Ray Sky with GALAXIS," UC Berkeley Astrophysics Department Lunch Talk, *Berkeley, CA, October 2025*
- "Galactic Axion Factories: Stellar Populations and X-Ray Signals for Axion Searches," Axions in Stockholm 2025, *Stockholm, Sweden, July 2025*
- "Galactic Axion Factories: Stellar Populations and X-Ray Signals for Axion Searches," The Strong CP Problem and its Possible Solutions 2025, *Pollica, Italy, June 2025*
- "Galactic Axion Factories: Stellar Populations and X-Ray Signals for Axion Searches," Berkeley 4D Seminar, *Berkeley, CA, April 2025*
- "Axion-Photon Sensitivity with NuSTAR Observations of M82 and M87," TeVPA 2024, *Chicago, IL, August 2024*
- "Axion-Photon Sensitivity with NuSTAR Observations of M82 and M87," DPF-PHENO 2024, *Pittsburgh, PA, May 2024*

- "Resurrecting Hitomi for X-ray Probes of Sterile Neutrino and Axion-Like-Particle Dark Matter with Blank Sky Observations," PASCOS 2023, *Irvine, CA, June 2023*

Collaboration Membership

- COSI, *Dark Matter Working Group* ..... 2025 -
- GALAXIS, *Dark Matter Working Group, Astrophysics Working Group* ..... 2025 -
- Mu2e, *Tracker Working Group* 2019-2022

Selected Awards, Honors, and Funding

- Sushant Singh Rajput Memorial Fellowship in Physics 2025
- Hearts to Humanity Eternal (H2H8) Graduate Research Grant (site) 2025
- Outstanding Graduate Student Instructor (OGSI) Award (site) ..... 2023 – 2024
- National Science Foundation Graduate Research Fellowship (site) ..... 2022 – 2025
- Percy Lionel Davis Award in Mathematics (site) 2022
- Cal Alumni Association's The Leadership Award and Scholarship (site) ..... 2021 – 2022, 2018 – 2019
- Astronaut Scholarship Foundation (ASF) Scholar (site) ..... 2021 – 2022

Teaching and Tutoring

Teaching Assistantships (GSI)

- Physics 110B: Classical Electrodynamics Part II *(instructor: Liang Dai)* ..... Fall 2023

Democratic Education at Cal (DeCal) | *Lead Course Facilitator and Teacher* site

- Astronomy 98: Beginner's Guide to the Universe *(faculty sponsor: Jessica Lu)* ..... Fall 2020 – Spring 2022

Department Peer Tutor and Group Study Facilitator | *Physics Department* site

- Physics 137A (Quantum Mechanics I) ..... Spring 2022
- Physics 110A (Electromagnetism and Optics I) ..... Fall 2021
- Physics 137B (Quantum Mechanics II) ..... Spring 2021, Fall 2020

Course Reader/Grader

- Physics 209: Classical Electromagnetism (Graduate Level) *(instructor: Liang Dai)* ..... Fall 2024
- Physics 137A: Quantum Mechanics I *(instructor: Swapan Chaterjee)* ..... Summer 2023
- Physics 110B: Classical Electrodynamics Part II *(instructor: Ehud Altman)* ..... Spring 2022
- Physics 112: Statistical Mechanics/Thermodynamics *(instructors: Michael Zaletel, Ehud Altman)* ..... Fall 2021
- Physics 137A: Quantum Mechanics I *(instructor: Dan Kasen)* ..... Spring 2021
- Physics 77: Introduction to Computational Techniques in Physics *(instructor: Yury Kolomensky)* ..... Fall 2020
- Math 54: Linear Algebra and Differential Equations *(instructor: Rui Wang)* ..... Summer 2020

Pedagogy Theory | *Courses Taken*

- MPS 375: Pedagogy in Math and the Physical Sciences ..... Spring 2023
- Student Learning Center (SLC) Astronomy and Physics Pedagogy Seminar ..... Fall 2021 – Spring 2022

Outreach, Service, and Mentoring

Berkeley Center for Theoretical Physics 4D Seminar site

- Organizer ..... Spring 2026

Astronomy on Tap San Francisco site

- Speaker, "From Astrophysics to Axions: The Glorious Hunt for Dark Matter" (Dec. 9) ..... Fall 2025

UC Berkeley Compass Lectures site

- Speaker, "The Cosmic Game of Hide-and-Seek: Chasing Dark Matter Across the Stars" (Apr. 18) ..... Spring 2025

UC Berkeley Astro Night Public Lecture site

- Speaker, "From Astrophysics to Axions: The Glorious Hunt for Dark Matter" (Oct. 3) ..... Fall 2024

Splash at Berkeley | *Administrator and Volunteer Teacher* site

- Administrator and Organizer ..... Spring 2024 –
- Class S1663: Cosmic Ghostbusters: Hunting the Universe's Invisible Beasts *(joint with Hannah Lykes)* Spring 2026
- Class S1570: Good Vibrations: Gravitational Waves and the Hidden Music of Spacetime ..... Fall 2025
- Class S1469: The Usual (Particle) Suspects: Culprits Beyond the Standard Model? ..... Spring 2025
- Class S1241: From Astrophysics to Axions: The Glorious Hunt for Dark Matter ..... Spring 2024, Fall 2024
- Class S919: So Long, and Thanks for All the Spacetime *(joint with Druv Punjabi)* ..... Spring 2022
- Class S838: Alge-cada-bra: The Algebraist's Answer to Life, the Universe, and Everything *(joint with Ryan Tamura)* ..... Fall 2021
- Rainstorm Virtual Class S732: The Quantum Complex *(joint with Ryan Tamura)* ..... Spring 2021
- Class S772: The Quantum Complex *(joint with Ryan Tamura)* ..... Spring 2021
- Class S667: Relativity and Spacetime ..... Fall 2020

Science Olympiad

- Science Olympiad at Berkeley: Exam Writer, Moderator, Grader – Solar System ..... Spring 2026, site
- Golden Gate Science Olympiad: Volunteer, Moderator & Grader – Astronomy ..... Spring 2025, site

UC Berkeley Major Curriculum Committee

- Graduate Representative ..... Fall 2022 – Spring 2023
- Undergraduate Representative ..... Fall 2021 – Spring 2022

MPS Scholars Mentoring Network | *Volunteer Mentor* site

- Mentor ..... Fall 2025 –

Berkeley Undergraduate Mathementoring Program (BUMP) | *Volunteer Mentor* site

- Mentor ..... Fall 2019 – Fall 2021

Society of Physics Students (SPS) Mentoring Program | *Volunteer Mentor* site

- Mentor ..... Fall 2020 – Spring 2022

Technical Skills

Languages and Software: Python, C/C++, ROOT, Linux/Unix, Bash, high-performance computing (HPC), Mathematica, LaTeX, Git, MESA, NSCool, CAMB/CosmoMC, Geant4, MEGAlib, Arduino, LabVIEW
Other: Electronics and Circuitry, Soldering